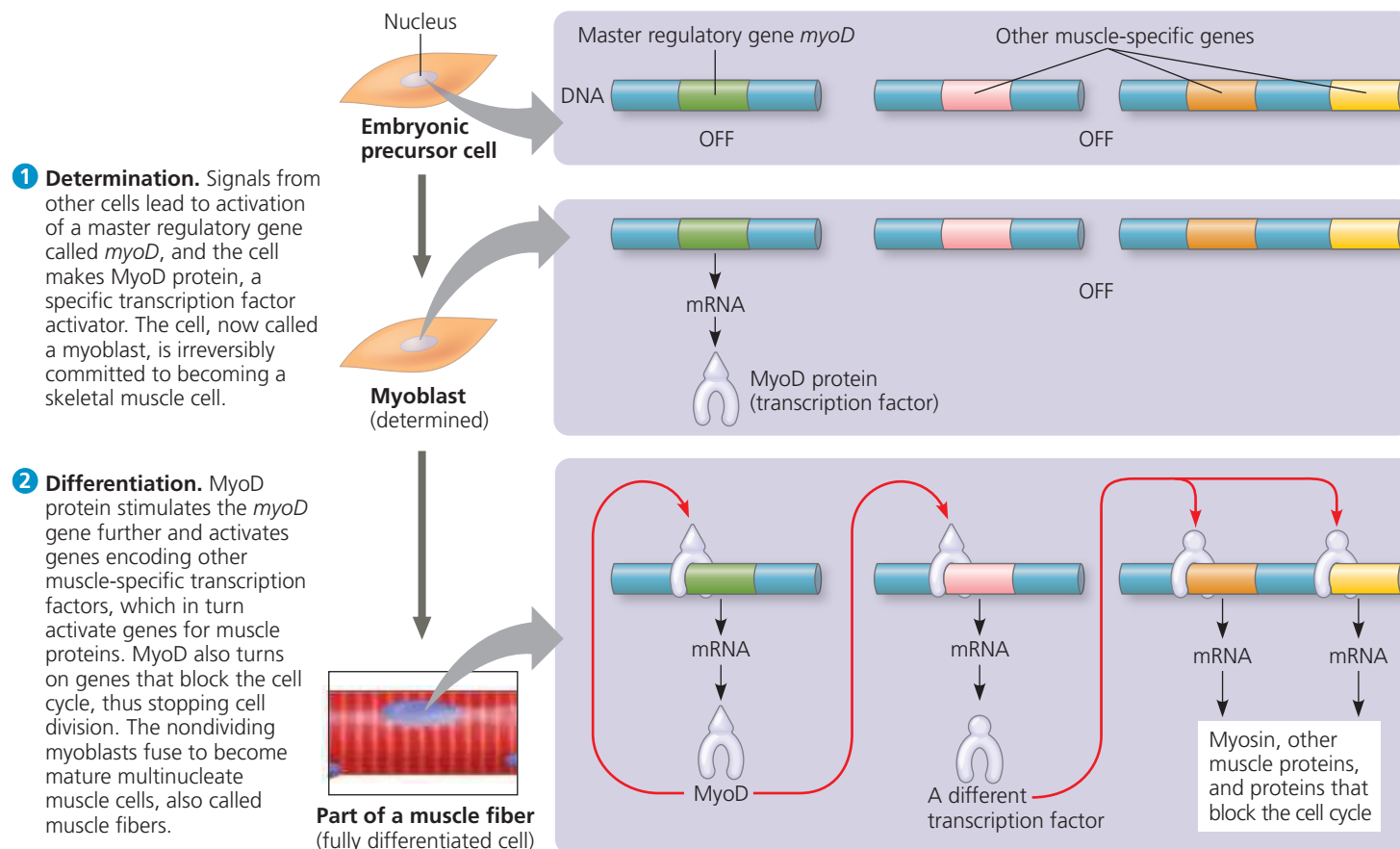


▼ **Figure 18.18 Determination and differentiation of muscle cells.** Skeletal muscle cells arise from embryonic cells as a result of changes in gene expression. (In this depiction, the process of gene activation is greatly simplified.)



WHAT IF? What would happen if a mutation in the *myoD* gene resulted in the production of an altered MyoD protein that could not activate the *myoD* gene?

the genes for proteins such as myosin and actin that confer the unique properties of skeletal muscle cells.

The determination and differentiation of other kinds of tissues may play out in a similar fashion. Experimental results support the idea that master regulatory proteins like MyoD might function by opening the chromatin in certain regions. This allows access to transcription machinery for activation of the next set of cell-type-specific genes.

Pattern Formation: Setting Up the Body Plan

We have now seen how different programs of gene expression that are activated in the fertilized egg can result in differentiated cells and tissues. But for the tissues to function effectively in the organism as a whole, the organism's *body plan*—its overall three-dimensional arrangement—must be established and superimposed on the differentiation process. Let's look at the molecular basis for the establishment of the body plan, using the well-studied fruit fly *Drosophila melanogaster* as an example.

Cytoplasmic determinants and inductive signals both contribute to spatially organizing the tissues and organs of an

organism in their characteristic places. This developmental process is referred to as **pattern formation**.

Just as the locations of the front, back, and sides of a new building are determined before construction begins, pattern formation in animals begins in the early embryo, when the major axes of an animal are established. In a bilaterally symmetrical animal, the relative positions of head and tail, right and left sides, and back and front—the three major body axes—are set up before the organs appear. The molecular cues that control pattern formation, collectively called **positional information**, are provided by cytoplasmic determinants and inductive signals (see Figure 18.17). These cues tell a cell its location relative to the body axes and to neighboring cells, and determine how the cell and its descendants will respond to future molecular signals.

During the early 20th century, classical embryologists made detailed anatomical observations of embryonic development in a number of species and performed experiments in which they manipulated embryonic tissues. Although this research laid the groundwork for understanding the mechanisms of development, it did not reveal the specific